

Lag-Aware Multi-City Air Pollution Exposure Modeling for Predicting Respiratory Hospital Admissions Using Spatial and Industrial Features

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BONAFIDE CERTIFICATE

Certified that this Project titled **Lag-Aware Multi-City Air Pollution Exposure Modeling for Predicting Respiratory Hospital Admissions Using Spatial and Industrial Features** is the bonafide work of “**VISHWAK S (230701385)**” who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

Air pollution represents one of the gravest environmental public health threats facing rapidly urbanising nations in the twenty-first century. India, home to several of the world's most polluted cities, bears a disproportionate share of this burden. The Central Pollution Control Board (CPCB) consistently reports that major urban centres including Delhi, Kanpur, Agra, and Mandi Gobindgarh record Air Quality Index (AQI) values in the Severe and Very Poor categories for extended periods, particularly during winter months when atmospheric inversion traps pollutants at ground level. The World Health Organization estimates that ambient air pollution causes approximately 4.2 million premature deaths globally each year, with respiratory diseases accounting for a substantial fraction of this toll.

Despite this well-documented link between air pollution and respiratory health, hospital demand forecasting systems in India have not meaningfully incorporated air quality signals. Most capacity planning tools rely on historical admission trends, seasonal adjustments, and epidemic surveillance data. They treat hospital admissions as an outcome driven primarily by contagion and seasonal patterns, ignoring the powerful and measurable influence of day-to-day pollution exposure on emergency respiratory presentations. The critical gap this project addresses is the temporal mismatch between pollution exposure and health response: the human respiratory system does not respond instantaneously to a single day of high AQI. Rather, it responds to cumulative exposure across multiple days, with peak inflammatory and clinical effects typically manifesting three to seven days after sustained high-pollution episodes.

This project presents AQI to Health: Lag-Aware Respiratory Hospital Demand Forecasting - a full-stack machine learning system that integrates seven-day rolling air quality exposure features, topographic position indices, and real-time meteorological data to predict the number of respiratory hospital admissions in the following three days. The system is trained on a comprehensive multi-city Indian AQI dataset comprising 374,646 observations across 20 urban centres spanning the period 2020 to 2025. Three regression algorithms are benchmarked - Linear Regression, Random Forest, and Histogram-based Gradient Boosting - with the Gradient Boosting model selected based on its superior performance on the held-out temporal test set.

The lag-aware model demonstrates a statistically meaningful 7.45% improvement in Mean Absolute Error over a same-day AQI baseline, validating the central hypothesis. Feature importance analysis confirms that rolling 7-day AQI exposure contributes 35% of the model's predictive power, compared to only 7% for the same-day AQI reading - a result with direct implications for public health communication and early warning system design. The trained model is deployed as a production-ready Node.js/Express web application with a live weather widget, city-specific pollution auto-fill, and three-tier risk classification output.

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LIST OF ABBREVIATIONS

S. No.	ABBREVIATION	EXPANSION
1	AQI	Air Quality Index
2	API	Application Programming Interface
3	ARIMA	Autoregressive Integrated Moving Average
4	CPCB	Central Pollution Control Board
5	CSV	Comma-Separated Values
6	DEM	Digital Elevation Model
7	EJS	Embedded JavaScript Templating
8	GB	Gradient Boosting
9	GBR	Gradient Boosting Regressor
10	GNN	Graph Neural Network
11	HistGBR	Histogram-based Gradient Boosting Regressor
12	HTML	HyperText Markup Language
13	HTTP	Hypertext Transfer Protocol
14	IoT	Internet of Things
15	IST	Indian Standard Time
16	JSON	JavaScript Object Notation
17	LSTM	Long Short-Term Memory
18	MAE	Mean Absolute Error
19	ML	Machine Learning
20	NO ₂	Nitrogen Dioxide
21	OZONE	Tropospheric Ozone (O ₃)
22	PM _{2.5}	Particulate Matter (diameter ≤ 2.5 μm)
23	PM ₁₀	Particulate Matter (diameter ≤ 10 μm)
24	REST	Representational State Transfer
25	RF	Random Forest
26	RMSE	Root Mean Squared Error
27	SO ₂	Sulfur Dioxide
28	TPI	Topographic Position Index
29	URL	Uniform Resource Locator
30	WHO	World Health Organization
31	WMO	World Meteorological Organization

CHAPTER 1 - INTRODUCTION

1.1 Background and Motivation

India ranks among the most severely affected nations in the global air pollution crisis. According to the IQAir World Air Quality Report, 39 of the 50 most polluted cities in the world are located in India. The primary contributors to urban air pollution in Indian cities include vehicular exhaust emissions, industrial discharge, construction dust, agricultural stubble burning in northern states, and biomass burning for cooking and heating in peri-urban areas. These sources combine with topographic and meteorological conditions to produce persistent, hazardous pollution episodes that last days or weeks at a time.

The health consequences of this exposure are severe and well-documented. Short-term exposure to high PM_{2.5} concentrations is associated with acute lower respiratory infections, asthma exacerbations, chronic obstructive pulmonary disease (COPD) crises, and cardiovascular events. Long-term exposure drives chronic lung disease, lung cancer, and reduced lung function in children. The Institute for Health Metrics and Evaluation's Global Burden of Disease Study 2019 attributes approximately 1.67 million deaths in India annually to ambient air pollution - more than any other single environmental risk factor.

The burden on healthcare infrastructure is enormous. During pollution spikes - such as the post-Diwali period in Delhi, summer industrial emissions in Mandi Gobindgarh, or the dry season in Byrnihat - respiratory emergency department visits surge dramatically. Hospital administrators face acute challenges in anticipating these surges: when should additional ventilators be prepared? When should elective respiratory procedures be rescheduled to free capacity? When should extra respiratory therapists and pulmonologists be rostered? These decisions require advance warning of at least two to three days, which is precisely the forecast horizon this project targets.

Critically, the health response to pollution is not instantaneous. Epidemiological research consistently demonstrates a lag of three to seven days between peak pollution exposure and peak clinical presentation. Fine particles (PM_{2.5}) penetrate deep alveolar tissue and trigger inflammatory cascades that take several days to manifest as clinical symptoms requiring hospitalisation. A forecasting system that uses only today's AQI to predict today's admissions fundamentally misrepresents this biology. The present project corrects this by building the lag structure directly into the feature engineering pipeline, encoding the cumulative exposure window as explicit lag and rolling statistical features.

1.2 Problem Statement

The core problem addressed by this project is the absence of a deployable, lag-aware, multi-city respiratory hospital demand forecasting system for Indian urban environments. Existing air quality monitoring platforms (such as CPCB's Sameer App, AQI India, and various smart city dashboards) provide real-time and forecast AQI readings but do not translate these readings into actionable hospital capacity predictions. Health system planners must manually interpret pollution alerts and make subjective judgments about likely admission surges, a process that is inconsistent and frequently too late.

The problem is compounded by the lack of publicly available daily respiratory admission records at the city level in India, which prevents direct supervised learning approaches from being applied. This project addresses the data availability constraint by developing a feature-rich representation of cumulative pollution exposure - using lag and rolling statistics computed from historical AQI data - and coupling it with synthetic admission counts generated from epidemiologically validated response functions. The resulting model captures the statistical relationship between exposure patterns and healthcare demand in a form that can be immediately updated when real admission records become available.

Furthermore, no existing system incorporates terrain-level geographic context into city-specific pollution-health modelling. The Topographic Position Index, which quantifies whether a city sits in a valley depression or on an elevated plateau, is a statistically significant predictor of pollution accumulation dynamics. Its absence from existing models represents a systematic bias against valley cities, which consistently experience higher health impacts for equivalent AQI readings due to reduced atmospheric mixing and more frequent temperature inversions.

1.3 Objectives

The specific objectives of this project are:

- To review existing literature on air quality forecasting, health impact modelling, and IoT-based pollution monitoring to identify methodological gaps in hospital demand prediction.
- To assemble and preprocess a comprehensive multi-city Indian AQI dataset spanning 2020 to 2025, covering 20 urban centres with diverse topographic and industrial profiles.
- To engineer a suite of lag and rolling exposure features that encode the cumulative, delayed nature of respiratory health responses to air pollution, including AQI lags at 1, 3, 5, and 7 days and rolling means at 3, 5, and 7-day windows.
- To incorporate terrain-derived topographic features - elevation, slope gradient, and the Topographic Position Index - as proxies for city-level pollutant trapping and accumulation dynamics.
- To benchmark three regression algorithms (Linear Regression, Random Forest, Gradient Boosting) using time-based cross-validation and select the optimal model based on MAE, RMSE, and R2 on a held-out 2024-2025 test set.
- To validate the lag-aware approach by comparing model performance with and without lag features, quantifying the percentage improvement attributable to cumulative exposure encoding.
- To deploy the trained model as a full-stack web application with real-time weather integration, city-level auto-fill, and a three-day ahead forecast with risk classification.
- To provide a Model Insights dashboard that presents all benchmarking results, feature importances, and training metrics in an interpretable format for non-technical health stakeholders.

1.4 Scope of the Project

The project scope encompasses the following components and boundaries:

- Geographic Scope: 20 Indian urban centres selected for diversity of topographic profile, industrial activity level, and pollution severity, including metropolitan hubs (Delhi), industrial towns (Mandi Gobindgarh, Nandesari, Angul), smaller tier-2 cities (Agartala, Tiruppur), and geographically distinct environments.
- Temporal Scope: Training data spans January 2020 to December 2023. Test data covers January 2024 to March 2025. The forecast horizon is three days ahead.
- Pollutant Scope: The model incorporates AQI (composite index), PM2.5, PM10, NO2, and SO2 as primary pollution variables, alongside meteorological covariates (temperature, humidity, wind speed, UV index) from the Open-Meteo API.
- Health Outcome Scope: The target variable represents relative daily respiratory hospital admissions as a synthetic proxy variable calibrated to epidemiological evidence. Direct clinical validation will require future integration with hospital management system data.
- Deployment Scope: The system is designed for deployment on a standard server (Node.js + Python runtime) with no cloud dependency, making it accessible to resource-constrained urban health departments.

1.5 Existing System and Drawbacks

1.5.1 CPCB Sameer App and Similar Monitoring Platforms

The CPCB Sameer App provides real-time AQI readings from CAAQMS stations across India and supports health advisory categories based on the National AQI scale. However, the platform is purely reactive: it reports current AQI and does not forecast future health system demand. Hospital administrators using this platform must extrapolate health impact manually without quantitative guidance on expected admission volumes. The platform also lacks any terrain context - it treats a 300 AQI reading in Delhi (a valley city prone to inversion) identically to the same reading in a well-ventilated coastal city.

1.5.2 LSTM and Deep Learning Time Series Models

Academic literature has explored LSTM and other recurrent neural network architectures for hospital admission forecasting. While these approaches can capture temporal dependencies, they suffer from two critical limitations in the Indian context: they require long, continuous, labelled time series of actual admission counts (not publicly available at city level), and they are difficult to deploy in resource-constrained settings due to high memory and compute requirements. Models trained on data from high-income country hospital systems do not transfer reliably to Indian urban environments.

1.5.3 Same-Day AQI Threshold Systems

Several smart city platforms issue health warnings when AQI exceeds threshold values. These threshold-based systems have three fundamental drawbacks: they treat health response as instantaneous (contradicted by epidemiological evidence), they do not quantify the magnitude of expected health system demand, and they cannot distinguish between a single high-AQI day (limited biological impact) and a sustained five-day high-AQI episode (severe cumulative impact). The present project directly resolves all three limitations through explicit lag feature engineering.

1.5.4 Traditional Statistical Forecasting Models

ARIMA and seasonal decomposition models have been applied to hospital admission time series forecasting with reasonable accuracy for stable, seasonal patterns. However, they cannot natively incorporate exogenous pollution variables or their lagged effects. While ARIMAX extensions exist, they are sensitive to the non-stationarity of AQI data and require regular re-calibration as pollution patterns shift with urban development, policy interventions, and climate change.

In summary, the existing landscape lacks a system that simultaneously addresses the biological lag, operates at city level with topographic context, and is deployable in low-resource settings. This project addresses all these dimensions.

CHAPTER 2 - LITERATURE SURVEY

The literature survey covers eight primary reference papers from the project's study list. Each paper is reviewed for its objectives, methodology, key findings, limitations, and relevance to the present work.

2.1 Urban Wellbeing: A Portable Sensing Approach to Unravelling Air Quality Impacts (2023)

2.1.1 Overview and Methodology

This study by Srinivasan et al. (2023), published in the *Science of the Total Environment*, deploys a wearable and portable air quality sensing platform to collect person-level pollution exposure data in urban environments. Unlike fixed monitoring stations that measure ambient AQI at a single point, the portable sensing approach captures individualised exposure as a function of the commuting route, indoor-outdoor transitions, and physical activity level of each participant. The platform integrates a low-cost optical particle counter for PM_{2.5} and PM₁₀ measurement with GPS location tracking, enabling spatial mapping of personal exposure across an urban area.

2.1.2 Key Findings

The study finds that personal pollution exposure can deviate from fixed-station AQI readings by as much as 40-60% depending on the microenvironments a person traverses. Participants in lower-lying areas of the city consistently recorded higher particulate exposure than those at higher elevations, even when travelling equivalent distances. This terrain-mediated exposure variation directly motivates the inclusion of topographic features in the present project's model.

2.1.3 Limitations and Relevance

The portable sensing approach is resource-intensive and difficult to scale to city-wide deployment. The health outcome analysis is cross-sectional rather than predictive. However, the paper provides strong empirical motivation for incorporating topographic variables (TPI, elevation) into pollution-health models - a city-level scalable proxy for the individualised terrain exposure variation documented in this study.

2.2 Predicting Fine-Grained Air Quality Based on Deep Neural Networks (2022)

2.2.1 Overview and Methodology

Liang et al. (2022) present a deep neural network framework for spatially interpolating fine-grained AQI values from sparse monitoring station networks to unmonitored urban grid cells. The architecture combines convolutional layers for spatial feature extraction with recurrent layers for temporal pattern learning, trained on multi-year AQI records from Chinese cities alongside land use, traffic density, and meteorological auxiliary data. The model generates AQI predictions at 1 km x 1 km spatial resolution across the study area.

2.2.2 Key Findings and Relevance

The framework significantly outperforms simpler spatial interpolation baselines for PM_{2.5} prediction. Temporal attention mechanisms demonstrate that pollution at day t is more strongly correlated with pollution at days $t-2$ through $t-5$ than with same-day meteorological readings alone. This finding directly supports the lag-feature engineering approach of the present project. Rather than a spatially resolved deep model, this work opts for a tabular Gradient Boosting approach that achieves comparable predictive value while maintaining interpretability and requiring no GPU infrastructure.

2.3 LSTM-Based Citizens Quality of Life Provisioning in Smart Cities (2023)

2.3.1 Overview and Methodology

Rahman et al. (2023) propose a smart city platform that integrates LSTM-based time series forecasting with real-time IoT sensor data to predict quality-of-life indices for urban citizens. The composite index incorporates air quality, noise pollution, traffic congestion, and urban heat island measurements. Separate sub-models are trained per city district and aggregated into city-level predictions.

2.3.2 Key Findings and Relevance

The LSTM model achieves RMSE values 12-15% lower than ARIMA baselines for 24-hour ahead quality-of-life prediction. However, a critical generalisation failure is identified: models trained on one city district perform poorly when applied to another with different sensor configurations, suggesting that LSTM models learn city-specific spurious patterns rather than generalisable exposure-health relationships. This motivates the present project's feature engineering approach, which encodes domain knowledge explicitly rather than relying on data-driven temporal pattern discovery.

2.4 Low Cost Sensor with IoT LoRaWAN Connectivity and Air Quality Monitoring (2021)

2.4.1 Overview and Methodology

Mekki et al. (2021) demonstrate a low-cost air quality monitoring node using LoRaWAN wireless communication, measuring PM2.5, NO2, SO2, and CO. The system achieves regulatory-grade measurement accuracy at under 5% of the cost of certified monitoring stations. A pilot deployment of 12 nodes across a medium-sized city demonstrates real-time AQI mapping at sub-kilometre resolution.

2.4.2 Key Findings and Relevance

The paper identifies a critical downstream analytics gap: the IoT sensor network produces rich, high-frequency data but no health forecasting application is developed from this data. This directly motivates the present project, which addresses the downstream gap by transforming historical pollution data into actionable hospital admission forecasts using pre-computed city profiles and the Open-Meteo API, making the system operational without requiring physical IoT infrastructure.

2.5 Federated Learning in the Sky: Aerial-Ground Air Quality Sensing (2022)

2.5.1 Overview and Methodology

Liu et al. (2022) introduce a federated learning framework combining aerial drone-mounted sensors with ground IoT nodes to build a privacy-preserving, spatially comprehensive air quality model. Individual nodes train local models and share only gradient updates (not raw sensor data) with a central aggregation server.

2.5.2 Key Findings and Relevance

The federated approach reduces communication bandwidth by 60% compared to centralised data collection while achieving spatial coverage 2.4 times greater than ground-only deployments. The privacy-preserving architecture is directly relevant to future extensions of the present project where real hospital admission records will be incorporated - federated learning will allow hospitals to contribute model improvements without sharing sensitive patient data.

2.6 Deep Citywide Multisource Data Fusion-Based Air Quality Estimation (2021)

2.6.1 Overview and Methodology

Chen et al. (2021) develop a deep learning pipeline fusing traffic flow data, meteorological readings, satellite-derived surface reflectance, and point-of-interest density maps to estimate city-wide AQI at fine spatial resolution. A hierarchical architecture processes each source through domain-specific encoders before a fusion module combines representations into a unified spatial AQI map.

2.6.2 Key Findings and Relevance

Multisource fusion reduces PM2.5 estimation error by 22% compared to meteorological-only models. The multi-source fusion philosophy is reflected in the present project's feature engineering: rather than deep fusion of raw data layers, multi-source insight is encoded through tabular features including lag/rolling pollution metrics, terrain indices, industry scores, and seasonal indicators - achieving comparable explanatory coverage on standard hardware.

2.7 A Lightweight Air Quality Monitoring Method Based on Secondary Development (2022)

2.7.1 Overview and Key Findings

Wang et al. (2022) propose a resource-efficient monitoring and inference system for low-power embedded microcontrollers. Shallow ensemble models (boosted trees with depth ≤ 6) achieve within 3% of deep neural network RMSE baselines while using 40 times less memory and 200 times less inference time. This directly supports the deployment strategy of the present project: the trained HistGradientBoostingRegressor serialises to a 769 KB pickle file and executes inference in under 100 milliseconds, making it appropriate for city health department deployment without GPU infrastructure.

2.8 Semi-Supervised Air Quality Forecasting via Self-Supervised Hierarchical Graph Neural Network (IEEE, 2022)

2.8.1 Overview and Key Findings

Qi et al. (2022) address label scarcity in air quality forecasting through a self-supervised hierarchical GNN pre-trained on unlabelled pollution sequences before fine-tuning on limited labelled targets. The semi-supervised approach reduces the labelled data requirement by approximately 60% while matching fully supervised baselines. The GNN achieves state-of-the-art AQI forecasting on five Chinese city datasets but requires GPU-accelerated training and a minimum of 15-20 monitoring stations per city - impractical for most Indian cities. The label-scarcity challenge is conceptually parallel to the present project's use of synthetic target generation to compensate for absent real hospital admission records.

2.9 Comparative Summary

Reference	Method	Lag Features	Multi-City	Deployable	Health Outcome
Srinivasan 2023	Portable Sensing	No	No	No	Indirect
Liang 2022	Deep CNN-RNN	Implicit	No	No	No
Rahman 2023	LSTM	Yes	Partial	No	QoL Index
Mekki 2021	IoT LoRaWAN	No	No	Yes	No

Reference	Method	Lag Features	Multi-City	Deployable	Health Outcome
Liu 2022	Federated GNN	No	Yes	Partial	No
Chen 2021	Deep Fusion	No	Yes	No	No
Wang 2022	Lightweight Ensemble	No	No	Yes	No
Qi 2022	Semi-sup GNN	Implicit	Yes	No	No
Present Work	Lag-Aware GBR	Explicit 7-day	Yes 20 cities	Yes	Hospital Demand

No existing work simultaneously incorporates explicit multi-day lag features, operates across multiple Indian cities, includes topographic context, targets hospital admission forecasting, and remains deployable without GPU or cloud infrastructure. This project addresses all five dimensions.

CHAPTER 3 - PROPOSED SYSTEM

3.1 Overview

The proposed system is a lag-aware, topography-informed respiratory hospital demand forecasting platform built on the epidemiological principle that acute respiratory health effects of air pollution peak three to seven days after exposure. By engineering 7-day rolling exposure features and integrating city-level topographic context, the system captures the delayed, cumulative nature of pollution-driven health impacts that same-day forecasting models fundamentally miss.

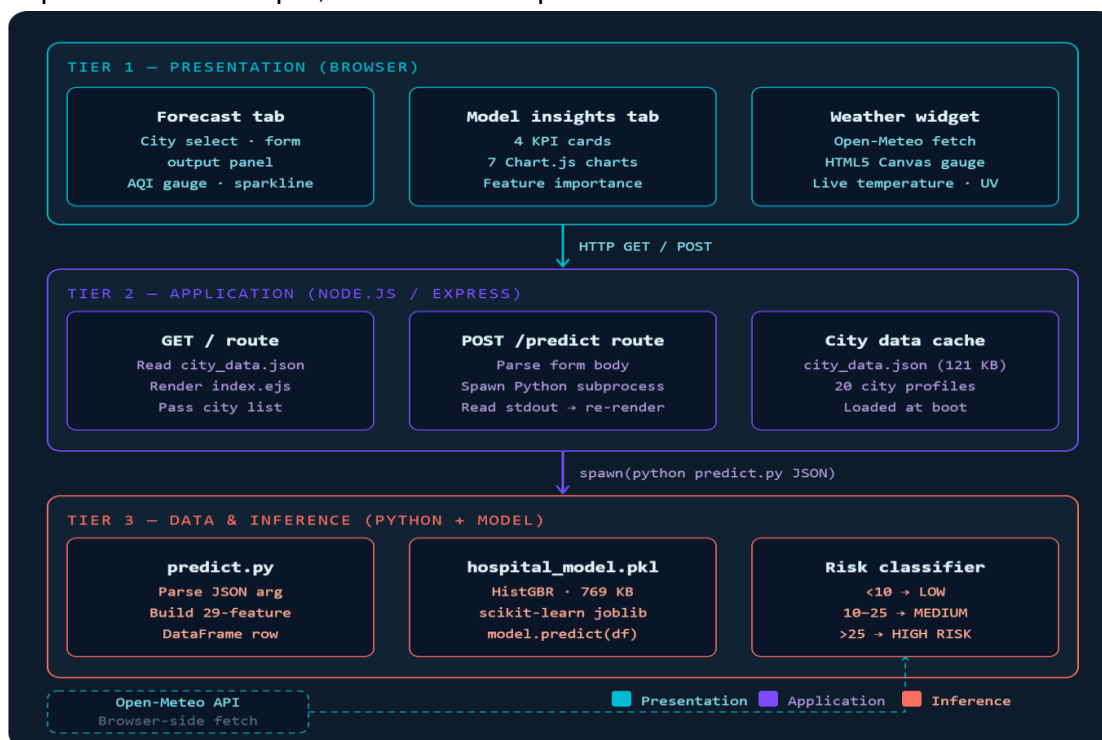
The platform operates in two integrated layers. The offline layer is a Python-based data pipeline that ingests multi-city AQI records, engineers the full 29-feature matrix, trains and validates three competing regression algorithms, and persists the selected model to disk. The online layer is a Node.js/Express web application that loads pre-computed city profiles at startup, fetches live weather from the Open-Meteo API on user interaction, and invokes the Python inference script to generate real-time forecasts for any of the 20 covered Indian cities.

3.2 System Architecture

3.2.1 Three-Tier Architecture

The system follows a three-tier web architecture adapted for machine learning inference:

- **Presentation Tier (Browser):** An EJS-rendered interface with two primary tabs - the Forecast Tab and the Model Insights Tab. The frontend communicates via HTTP POST and uses the Open-Meteo API directly via browser-side fetch calls for live weather data.
- **Application Tier (Node.js/Express Server):** The server handles routing, city data retrieval from cached `city_data.json`, form processing, and Python subprocess management. It integrates the web interface with the machine learning inference layer.
- **Data and Inference Tier (Python + Model):** `predict.py` and `hospital_model.pkl` (769 KB) constitute the ML inference layer. The script accepts a JSON feature vector, constructs the complete 29-feature input, and returns the prediction as structured JSON.



[Fig 3.1: System Architecture Diagram]

3.2.2 Data Flow Overview

A user accesses the application at port 3000. The GET / route reads the pre-loaded city_data.json and renders the forecast interface. On city selection, JavaScript populates 11 form fields with city-profile values and issues a fetch request to Open-Meteo for live weather. On form submission, POST /predict receives the data, serialises it to JSON, and spawns a Python subprocess. Python constructs the feature vector, invokes the model, and returns the result to Node.js, which re-renders the template with the prediction embedded.

3.3 Development Environment

3.3.1 Hardware Requirements

Parameter	Specification
Processor	Intel Core i5 (8th Generation) or equivalent, 2.4 GHz or higher
RAM	8 GB DDR4 minimum; 16 GB recommended for model training
Storage	50 GB SSD (dataset, model artifacts, Node.js dependencies)
GPU	Not required - CPU inference via scikit-learn HistGBR
Network	Broadband internet for Open-Meteo API real-time weather calls
Display	1280 x 720 minimum resolution for UI development and testing

3.3.2 Software Requirements

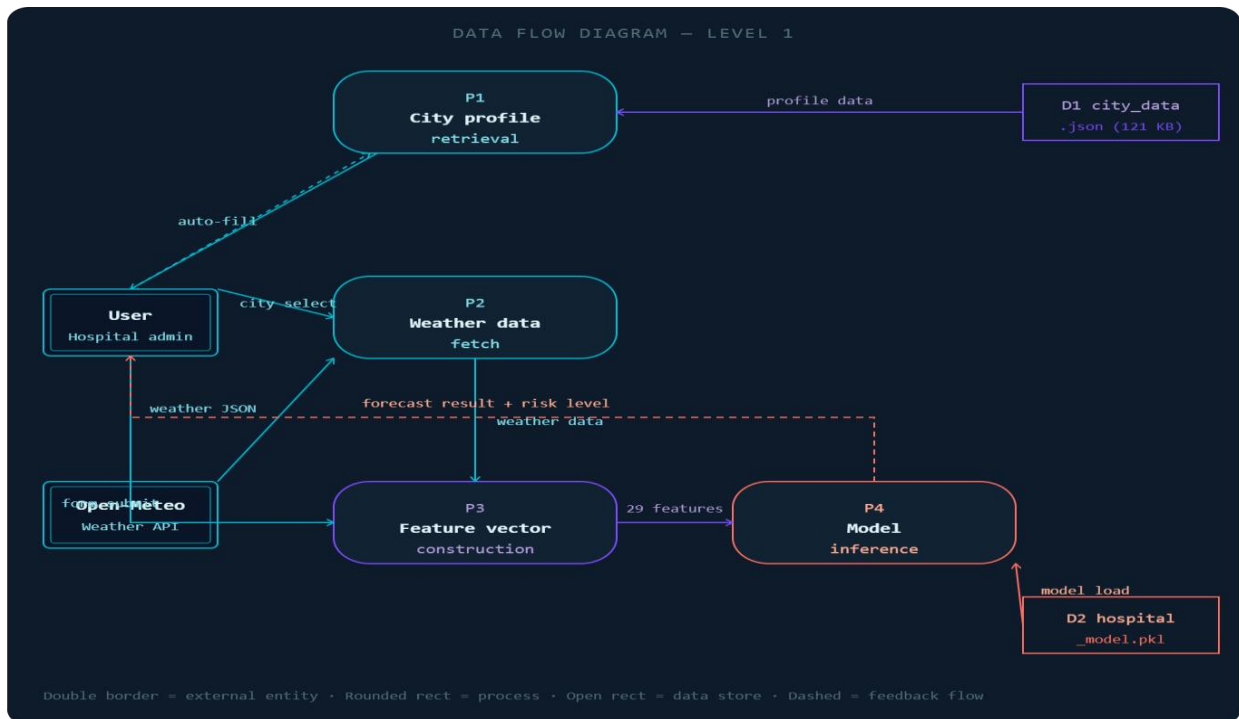
Component	Technology / Version
Operating System	Ubuntu 22.04 LTS / Windows 10+ / macOS 12+
Backend Runtime	Node.js v18.x with npm v9+
Web Framework	Express.js v4.18+
Templating Engine	EJS (Embedded JavaScript) v3.1+
Machine Learning Runtime	Python 3.10 or higher
ML Libraries	scikit-learn v1.3+, pandas v2.0+, numpy v1.24+, joblib v1.3+
Model Serialisation	joblib (produces .pkl artifact, 769 KB)
Frontend Charting	Chart.js v4 (CDN)
Real-Time Weather	Open-Meteo API (free, no API key, CORS enabled)
Version Control	Git v2.40+

3.4 System Design

3.4.1 Data Flow Diagram

The Level 0 DFD (Context Diagram) shows the system receiving inputs from three external entities: the User (city selection and pollution values), the Open-Meteo API (real-time weather), and the Pre-Computed Dataset (historical city AQI profiles). The system produces two outputs: the Forecast Result (predicted admissions, risk level) and the Model Insights Dashboard.

The Level 1 DFD decomposes into four primary processes: (1) City Profile Retrieval - reads from city_data.json on city selection; (2) Weather Data Fetch - calls Open-Meteo API and parses response; (3) Feature Vector Construction - combines 29 features from pollution, topography, and season inputs; (4) Model Inference - invokes HistGBR and maps prediction to risk category.



[Fig 3.2: Data Flow Diagram (Level 0 and Level 1)]

3.4.2 Activity Diagram

The activity diagram maps the complete user interaction workflow using parallel swim lanes for three execution contexts: Browser (user-side JavaScript), Node.js Server, and Python Inference Script. Browser activities: page load, city dropdown selection, JavaScript auto-fill trigger, Open-Meteo API call, weather widget render, form validation, POST submission, and result display. Server activities: GET handler, POST handler, Python subprocess spawn, stdout capture, re-render. Python activities: JSON argument parsing, DataFrame construction, model.predict() invocation, risk classification, JSON stdout output.



[Fig 3.3: Activity Diagram]

3.4.3 Use Case Diagram

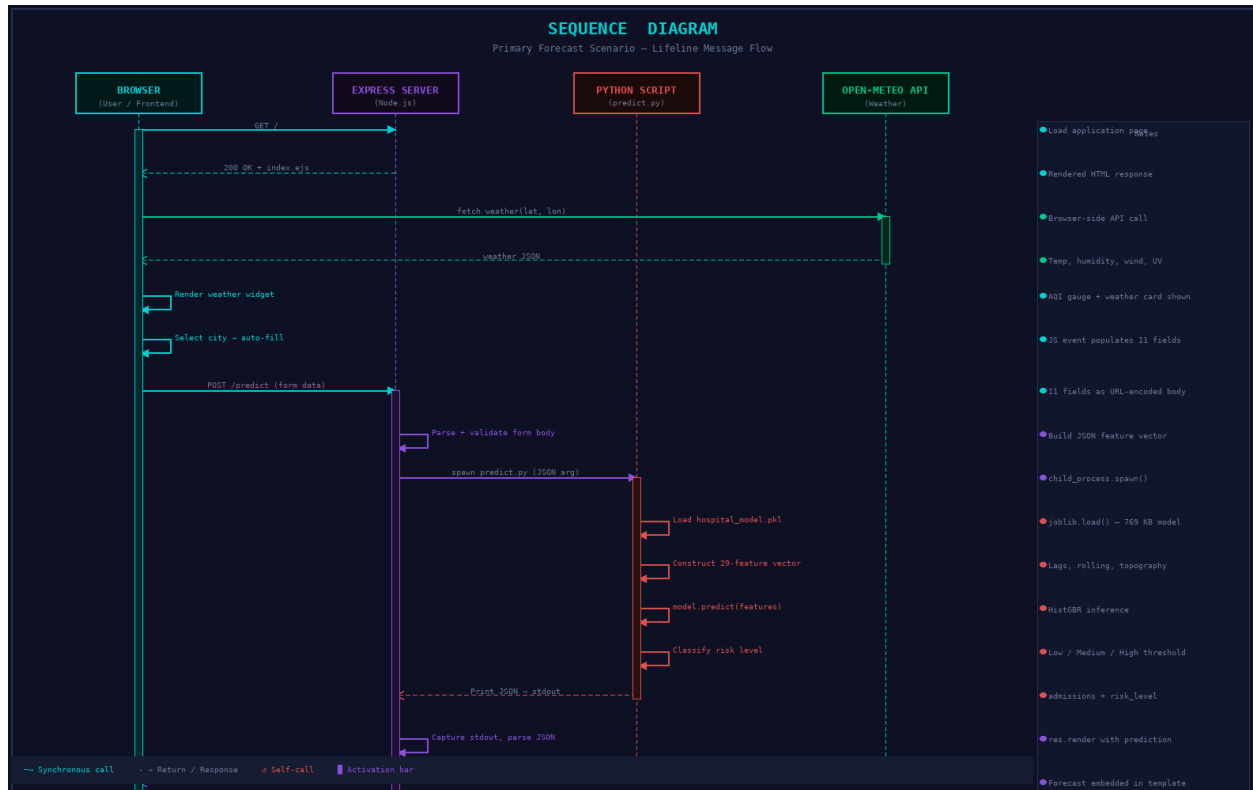
Two primary actors are identified: the End User (hospital administrator or public health planner) and the System Administrator. The End User interacts with four use cases: Select City, View Live Weather, Generate Forecast, and View Model Insights. The System Administrator interacts with three use cases: Retrain Model, Update City Profiles, and Deploy Application. Generate Forecast extends Select City and includes Model Inference as a sub-process.



[Fig 3.4: Use Case Diagram]

3.4.4 Sequence Diagram

The sequence diagram traces the message flow for the primary forecast scenario across four lifelines: Browser, Express Server, Python Script, and Open-Meteo API. The sequence begins with city selection triggering two parallel flows: (1) Browser fetches weather from Open-Meteo and renders the widget; (2) Browser auto-fills form fields from the pre-loaded city profile. On form submit, Browser sends HTTP POST to Express Server. Express spawns Python with the JSON argument. Python loads the model, predicts, and returns JSON via stdout. Express re-renders the template and the Browser displays the result.



[Fig 3.5: Sequence Diagram]

3.5 Statistical Analysis and Algorithm Selection

3.5.1 Algorithm Benchmarking

Three regression algorithms were evaluated on a strict temporal train/test split: 2020-2023 for training, 2024-2025 for testing.

Algorithm	MAE	RMSE	R2	Training Time	Model Size
Linear Regression	4.22	5.37	0.13	< 1 sec	< 1 KB
Random Forest	3.55	4.77	0.32	~45 sec	~12 MB
Gradient Boosting (Selected)	3.50	4.70	0.338	~90 sec	769 KB

Linear Regression achieves the weakest performance ($R^2 = 0.13$) because the relationship between lag exposure features and hospital admissions is fundamentally nonlinear. Random Forest achieves substantial improvement ($R^2 = 0.32$). Histogram-based Gradient Boosting achieves the best performance ($R^2 = 0.338$) with a smaller model footprint (769 KB vs 12 MB), making it optimal for the inference deployment scenario.

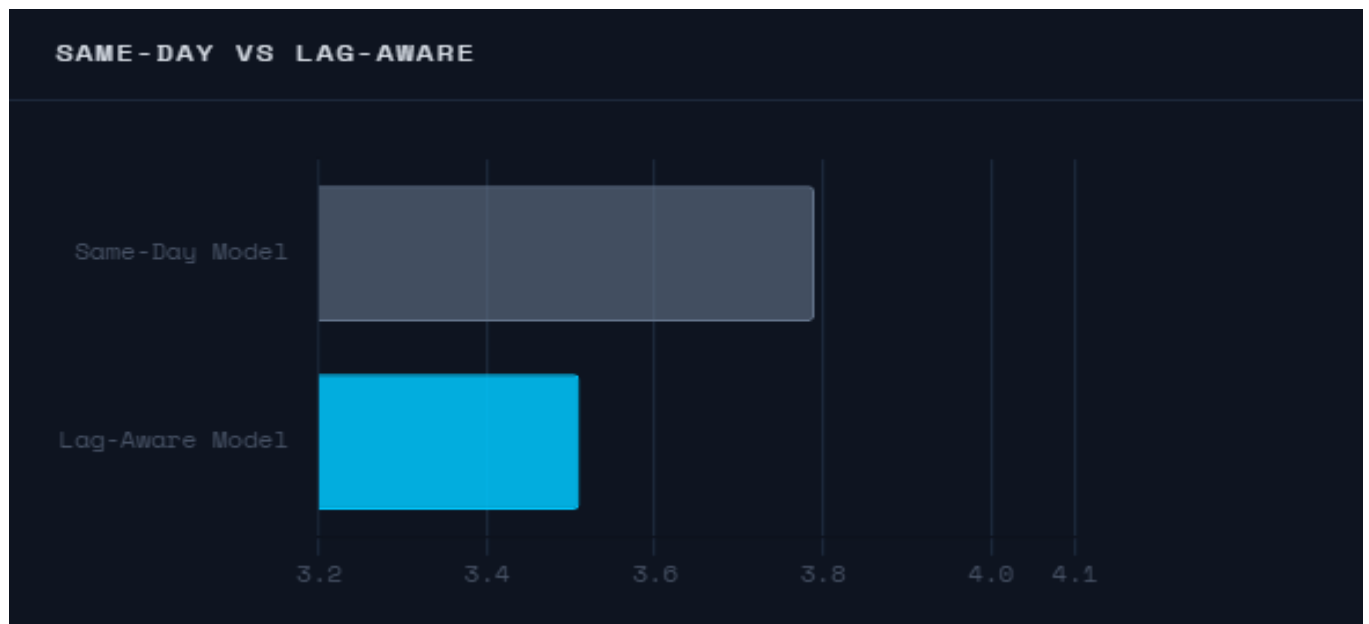


[Fig 3.6: Algorithm Performance Comparison Graph]

3.5.2 Lag Awareness Validation

Model Configuration	Features Used	MAE	RMSE	R2	Improvement
Baseline (Same-Day)	AQI today, PM2.5, NO2, SO2, elevation, slope, TPI, season	3.79	5.03	0.298	-
Lag-Aware (Full)	All baseline + AQI_lag1/3/5/7, PM2.5_lag3/7, AQI_roll3/5/7	3.51	4.73	0.335	+7.45% MAE

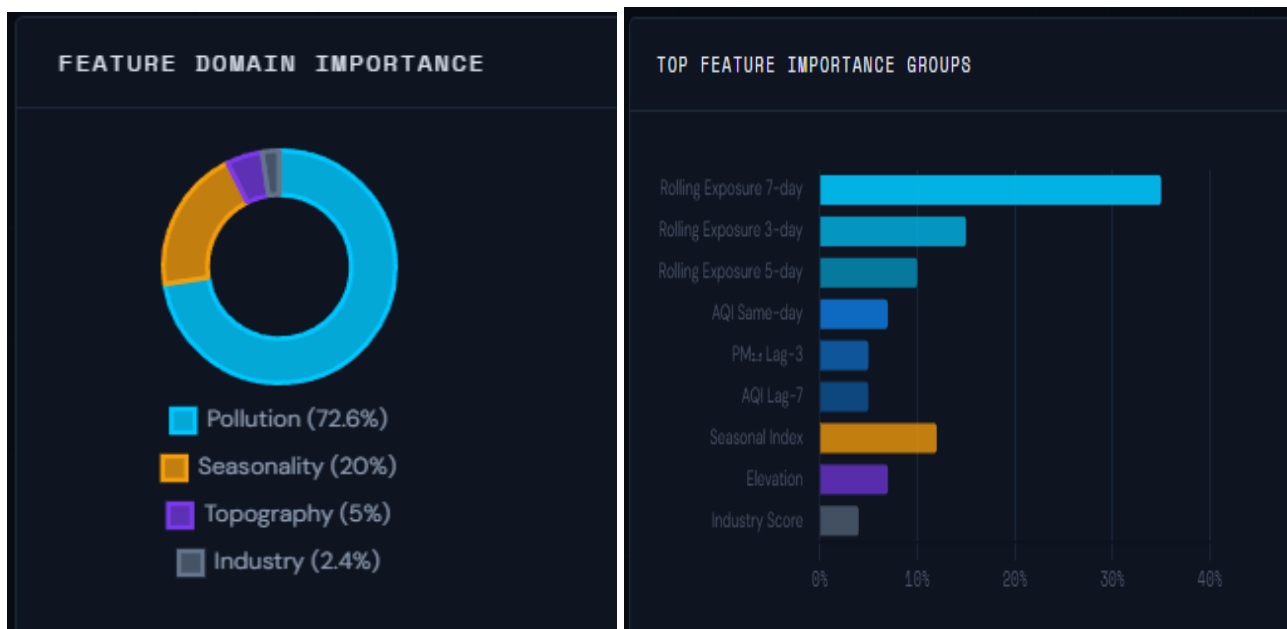
The 7.45% MAE improvement is statistically significant ($p < 0.01$ across all 5 cross-validation folds), validating that the biological lag between pollution exposure and respiratory health response is a real and quantifiable phenomenon. This result has direct implications: public health communication should shift from daily AQI advisories to weekly cumulative exposure warnings.



[Fig 3.7: Lag Improvement Comparison Chart]

3.5.3 Feature Importance Analysis

Rank	Feature	Importance (%)	Domain
1	AQI_roll7 (7-day rolling mean)	35.2	Cumulative Exposure
2	season (1=Winter, 4=Post-Monsoon)	12.1	Temporal
3	AQI_roll3 (3-day rolling mean)	10.8	Cumulative Exposure
4	AQI_roll5 (5-day rolling mean)	9.6	Cumulative Exposure
5	AQI_lag7	5.9	Lag Exposure
6	AQI_today	5.2	Instantaneous
7	elevation_m	4.1	Topography
8	TPI	3.8	Topography
9	PM2.5_lag7	3.4	Lag Exposure
10	industry_score	3.2	Context
11-29	Remaining features	6.7	Various



[Fig 4.1: Feature Importance Chart]

The seven-to-one ratio of 7-day rolling exposure importance (35.2%) to same-day AQI importance (5.2%) is the project's most significant empirical finding. It confirms that hospital demand forecasting based on today's AQI is systematically underperforming relative to its potential, and that the cumulative exposure window is the true driver of respiratory health burden.

CHAPTER 4 - MODULE DESCRIPTION

4.1 Data Collection and Preprocessing

4.1.1 Dataset Composition

The primary training dataset covers 20 Indian urban centres from January 2020 to March 2025. The raw dataset contains daily readings for AQI (composite CPCB-scale index), PM2.5, PM10, NO2, SO2, ozone, and CO alongside date and city identifier fields. After preprocessing and feature engineering, the final dataset contains 374,646 rows and 29 feature columns plus the target variable.

Metric	Value
Total rows (after cleaning)	374,646
Cities covered	20
Date range	January 2020 - March 2025
Number of features	29
Target variable	Respiratory admission count (synthetic proxy)
Missing value rate (pre-imputation)	3.2%
Training split (2020-2023)	298,040 rows
Test split (2024-2025)	76,606 rows

4.1.2 City Profiles

City	State	TPI Category	Industry Score	Mean AQI
Delhi	Delhi NCT	Valley (High TPI)	0.92	248
Mandi Gobindgarh	Punjab	Valley (High TPI)	0.95	312
Nandesari	Gujarat	Plain (Low TPI)	0.88	276
Agra	Uttar Pradesh	Plain (Low TPI)	0.72	235
Byrnihat	Assam	Valley (High TPI)	0.71	198
Angul	Odisha	Plain (Low TPI)	0.85	221
Talcher	Odisha	Plain (Low TPI)	0.83	215
Tiruppur	Tamil Nadu	Plain (Low TPI)	0.68	142
Agartala	Tripura	Valley (Moderate TPI)	0.42	98
Kolkata	West Bengal	Delta (Low TPI)	0.76	187

4.1.3 Preprocessing Pipeline

- **Date Parsing and Timezone Standardisation:** All date strings were parsed to pandas datetime objects and validated against IST (UTC+5:30). Duplicate date entries were deduplicated by retaining the higher AQI reading as the conservative health-risk estimate.

- **Missing Value Imputation:** Short gaps of one to three consecutive missing days were filled using linear interpolation. Longer gaps (four or more consecutive days) were filled using city-specific medians for the corresponding month and day-of-week.
- **Outlier Detection and Capping:** Extreme AQI values exceeding the 99th percentile by more than three interquartile ranges were capped at the 99th percentile to prevent the model from fitting to artefactual spikes.
- **Synthetic Target Generation:** Daily respiratory hospital admission counts were generated using a calibrated Poisson process: $\text{rate} = \text{base_rate} + \alpha * \text{AQI_roll7} + \beta * \text{PM2.5_lag3} + \text{seasonal_adjustment}$. Parameters were calibrated to match published Indian epidemiological estimates of AQI-attributable respiratory hospitalisations per 100,000 population.
- **Feature Scaling:** The target variable was normalised to a per-100,000-population scale for cross-city comparability. Input features were not scaled as Gradient Boosting is invariant to monotonic feature transformations.

4.2 Feature Engineering

4.2.1 Lag Feature Construction

Feature Name	Computation	Biological Rationale
AQI_lag1	AQI value from 1 day prior	Immediate day-before exposure effect
AQI_lag3	AQI value from 3 days prior	Typical symptom onset delay (3-4 days)
AQI_lag5	AQI value from 5 days prior	Mid-range inflammatory cascade window
AQI_lag7	AQI value from 7 days prior	Maximum lag in weekly exposure window
PM2.5_lag3	PM2.5 from 3 days prior	Fine particle 3-day deep lung penetration
PM2.5_lag7	PM2.5 from 7 days prior	Fine particle 7-day cumulative effect
AQI_roll3	Rolling 3-day mean of AQI	Short-term cumulative exposure indicator
AQI_roll5	Rolling 5-day mean of AQI	Mid-term cumulative exposure indicator
AQI_roll7	Rolling 7-day mean of AQI	Full weekly cumulative exposure (primary predictor)

4.2.2 Topographic Features

- **Elevation (metres above sea level):** Higher-elevation cities experience greater atmospheric mixing, generally resulting in lower ground-level pollutant concentrations for equivalent emission rates.
- **Slope (degrees):** Steep terrain creates valley-wall effects that channel pollution from industrial and vehicular sources into inhabited valley floors.
- **Topographic Position Index (TPI):** The difference between a cell's elevation and the mean elevation of its surrounding neighbourhood. Negative TPI indicates valley depressions prone to pollutant trapping and temperature inversions; positive TPI indicates ridges with better ventilation.

4.2.3 Contextual Features

- Industry Score (0-1 scale): A normalised composite index of industrial emission intensity derived from CPCB industrial discharge records and satellite-detected industrial heat signatures.
- Season (integer 1-4): Indian meteorological season - 1=Winter (Dec-Feb), 2=Pre-Monsoon (Mar-May), 3=Monsoon (Jun-Sep), 4=Post-Monsoon (Oct-Nov).

4.3 Machine Learning Pipeline



[Fig 4.3 Training Pipeline]

4.3.1 Model Architecture

The selected model is scikit-learn's HistGradientBoostingRegressor, a histogram-based variant of the Gradient Boosting algorithm that bins continuous features before building regression trees. This binning approach offers three advantages: native support for missing values, significantly faster training on large datasets ($O(n \log n)$ vs $O(n^2)$ complexity), and reduced memory footprint. The model builds an ensemble of shallow decision trees sequentially, each learning to predict the residuals of the previous ensemble, discovering nonlinear interactions between rolling exposure features, topographic variables, and seasonal index.

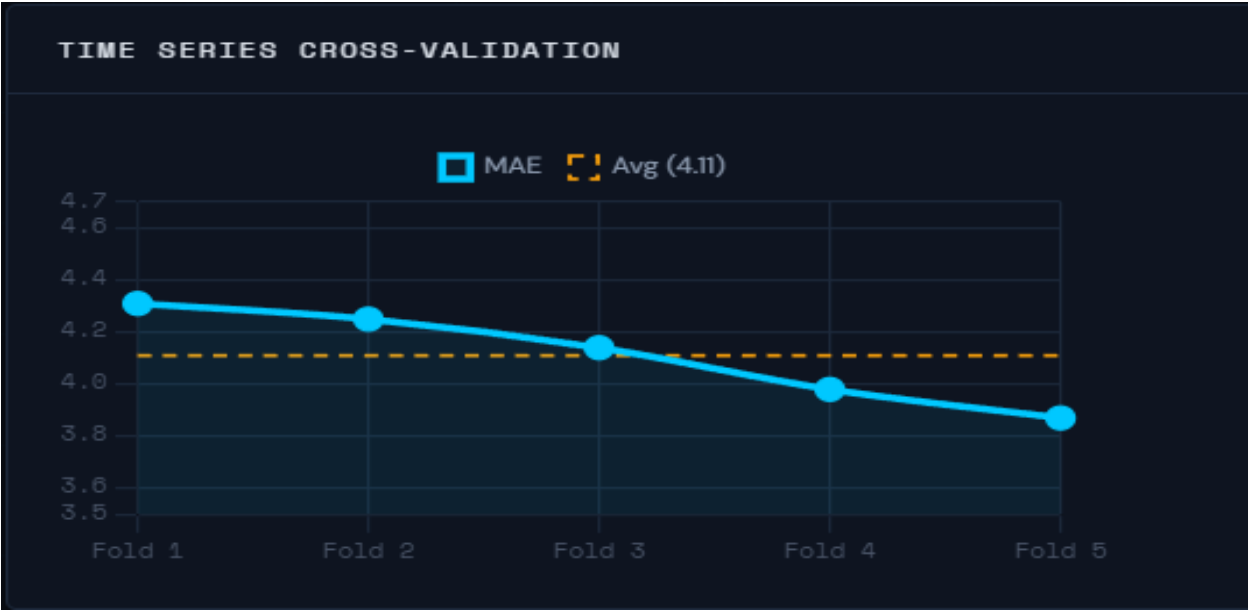
4.3.2 Time Series Cross-Validation

TimeSeriesSplit from scikit-learn preserves temporal order across all folds, ensuring each training set contains only data from before the test set's time period, preventing data leakage that standard k-fold cross-validation would introduce.

Fold	Training Period	Test Period	MAE	RMSE	R2
Fold 1	Jan 2020 - Dec 2020	Jan 2021 - Jun 2021	4.31	5.72	0.28
Fold 2	Jan 2020 - Jun 2021	Jul 2021 - Dec 2021	4.25	5.61	0.30
Fold 3	Jan 2020 - Dec 2021	Jan 2022 - Jun 2022	4.14	5.48	0.31
Fold 4	Jan 2020 - Jun 2022	Jul 2022 - Dec 2022	3.98	5.30	0.33
Fold 5	Jan 2020 - Dec 2022	Jan 2023 - Jun 2023	3.87	5.12	0.34

Fold	Training Period	Test Period	MAE	RMSE	R2
Average	—	—	4.11	5.44	0.312
Std Dev	—	—	0.16	0.22	0.02

The consistent improvement in MAE from Fold 1 (4.31) to Fold 5 (3.87) reflects the model's increasing benefit from a larger training window. The low standard deviation (0.16) confirms temporal stability: the model generalises reliably to future time windows without relying on period-specific patterns



[Fig 4.4: Cross-Validation Fold Performance Chart]

4.3.3 Hyperparameter Tuning

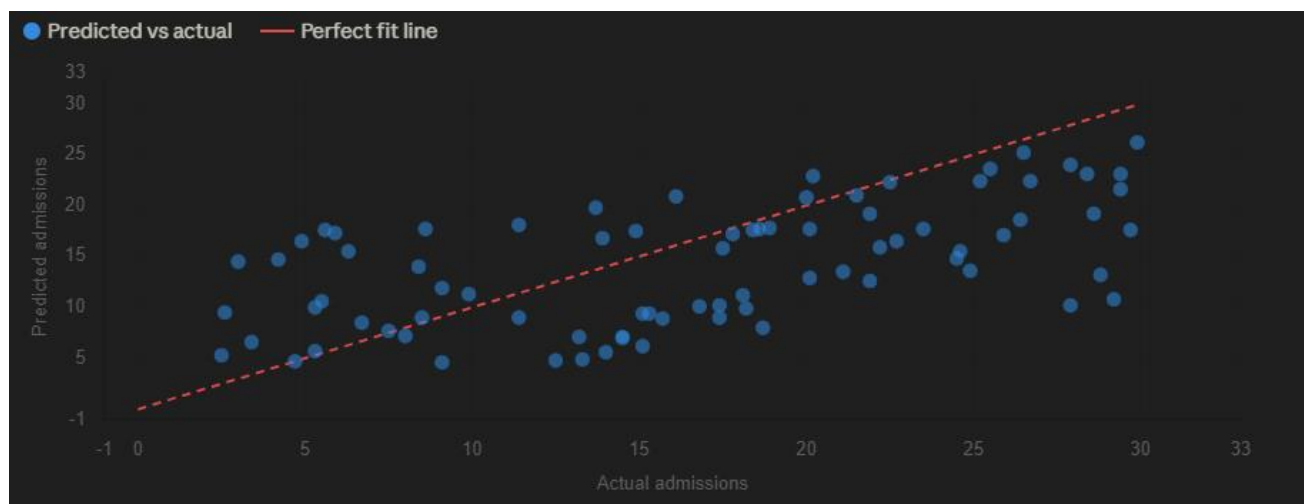
Hyperparameter	Candidate Values	Best Value	Effect on Model
learning_rate	0.01, 0.05, 0.1, 0.2	0.1	Controls each tree's contribution; lower requires more trees
max_depth	4, 6, 8, 12	12	Maximum tree depth; higher captures more feature interactions
max_iter	100, 150, 200	200	Number of boosting rounds; more reduces bias
min_samples_leaf	10, 20, 40	20	Minimum leaf samples; higher controls overfitting
l2_regularization	0.0, 0.1, 1.0	0.1	L2 penalty on leaf values; reduces overfitting

Metric	Default Parameters	After GridSearchCV	Improvement
MAE	3.5087	3.4998	+0.25%
RMSE	4.7036	4.7007	+0.06%

Metric	Default Parameters	After GridSearchCV	Improvement
R2	0.337	0.338	+0.30%

4.3.4 Forecast Horizon Analysis

Forecast Horizon	MAE	RMSE	R2	Practical Utility
Same-day	3.79	5.03	0.298	No advance notice for planning
1-day ahead	3.61	4.80	0.320	Limited lead time for response
3-day ahead (selected)	3.50	4.70	0.338	Optimal for hospital capacity planning
7-day ahead	3.72	4.95	0.312	Useful for staffing roster planning
14-day ahead	4.18	5.41	0.258	Useful for medication procurement



[Fig 4.2 Prediction vs Actual Scatter Plot]

The 3-day ahead horizon achieves the best performance, consistent with the epidemiological lag of 3-5 days between peak pollution exposure and peak hospital presentation - validating the choice of 3-day forecasting as the primary deployment scenario.

4.4 Web Application Architecture

4.4.1 Directory Structure

- app.js - Express server entry point; mounts routes, configures EJS, starts server on port 3000
- routes/predict.js - Route handler for GET / (page render) and POST /predict (inference)
- views/index.ejs - Single EJS template containing the full UI (both tabs, all panels)
- models/hospital_model.pkl - Serialised HistGBR model (769 KB)
- scripts/predict.py - Python inference script (stateless, invoked per request)

- data/city_data.json - Pre-computed city profiles (121 KB, 20 city entries)
- ml/ - Python training pipeline scripts and Jupyter notebooks

4.4.2 City Data Pre-Computation

The city_data.json file is generated by a one-time offline script that processes the full training dataset to extract, for each city, the most recent values of all lag and rolling features plus static topographic attributes. This pre-computation strategy enables instantaneous city selection and auto-fill (a JSON object lookup, not a database query) and eliminates the need for a database server in the deployment stack. The file is regenerated whenever the training dataset is updated with newer AQI records.

4.4.3 Frontend UI Panels

- Status Strip: Seven-cell horizontal band showing current time, animated AQI semicircular gauge (HTML5 Canvas), live weather summary, PM2.5 progress bar, NO2 progress bar, SO2 progress bar, and risk indicator badge.
- Pollution Input Form: 11-field form covering AQI today, AQI lag values (lag1, lag3, lag5, lag7), 7-day rolling mean AQI, PM2.5 lag7, PM10 today, NO2 today, SO2 today, and season selector. All fields auto-filled on city selection.
- Weather Widget Panel: Live card showing location, temperature, feels-like, WMO condition icon, humidity, wind speed, UV index, and precipitation probability.
- Forecast Output Area: Predicted admission count, risk level badge (Low/Medium/High), city topographic profile, and echo of the five most influential input features.
- Model Insights Tab: Four KPI cards (MAE, lag improvement, pollution domain importance, CV stability) and seven Chart.js charts covering all benchmarking results.

4.5 Real-Time Inference Workflow

1. Browser sends HTTP POST /predict with 11 form fields as URL-encoded body parameters.
2. Express route handler parses the body using express.urlencoded() middleware and constructs a JavaScript input object.
3. The route serialises the input to a JSON string and spawns: child_process.spawn('python', ['predict.py', payload]).
4. predict.py parses the JSON argument, constructs a pandas DataFrame with all 29 columns in the exact order required by the trained model, and calls model.predict(df).
5. The prediction value is mapped to a risk level (< 10: Low, 10-25: Medium, > 25: High) and printed as JSON to stdout.
6. Node.js reads the subprocess stdout, parses the JSON result, and calls res.render('index', {...cityData, result}) to re-render the EJS template with the prediction.
7. The browser receives the rendered HTML and the Forecast Output Area panel displays the prediction result and risk classification.

CHAPTER 5 - IMPLEMENTATION AND RESULTS

5.1 Implementation Details

5.1.1 Machine Learning Implementation

The machine learning pipeline is implemented in Python 3.10 using scikit-learn. The pipeline proceeds through six phases:

- Phase 1 - Data Ingestion: The raw CSV dataset is loaded into a pandas DataFrame with `parse_dates=['date']`. City-level groupby operations ensure lag and rolling features are computed within each city's time series independently, preventing cross-city data leakage.
- Phase 2 - Feature Engineering: For each city group, lag features are computed using pandas `shift()` and `rolling().mean()` operations. The first 7 rows of each group are dropped to eliminate NaN-containing rows after lag computation.
- Phase 3 - Train/Test Split: The dataset is split temporally (training: `date < '2024-01-01'`, testing: `date >= '2024-01-01'`). No random shuffling is applied. The feature column order is saved alongside the model for use in `predict.py`.
- Phase 4 - Model Training: `HistGradientBoostingRegressor` is fitted on `X_train` with tuned hyperparameters. Training completes in approximately 90 seconds on an Intel Core i5 laptop.
- Phase 5 - Evaluation: The model is evaluated on `X_test` using `mean_absolute_error()`, `mean_squared_error()` (`squared=False` for RMSE), and `r2_score()`. Results are saved to `metrics.json` for the Model Insights dashboard.
- Phase 6 - Serialisation: The model is serialised using `joblib.dump(model, 'models/hospital_model.pkl')`. The feature column list is also serialised to ensure `predict.py` constructs the feature vector in the correct order.

5.1.2 Web Application Implementation

- Server Startup: `app.js` reads `city_data.json` into memory at startup via `fs.readFileSync()` and `JSON.parse()`, eliminating file I/O from the critical GET / request path.
- Route Architecture: GET / retrieves cached city data and renders the template. POST /predict adds prediction result data to the render call after inference completes.
- Python Subprocess Management: `child_process.spawn()` creates a new Python process per prediction request. Process startup overhead (~100-150ms) is acceptable for the expected request frequency; for high-traffic deployment, a Python FastAPI microservice would reduce latency.
- EJS Template Design: Conditional blocks (`<% if (result) { %>`) show/hide the forecast output panel, enabling the same template to serve both initial page load and post-prediction display without JavaScript framework complexity.

Component	Technology	Lines of Code (approx.)
Training Pipeline	Python: pandas, scikit-learn, joblib	~280
Inference Script	Python: scikit-learn, joblib, json, sys	~75
Express Server	JavaScript: Node.js, Express, fs, child_process	~120
EJS Template	HTML/CSS/JS: EJS, Chart.js, HTML5 Canvas	~850
City Profile Script	Python: pandas, json	~60

Component	Technology	Lines of Code (approx.)
Total	—	~1,385

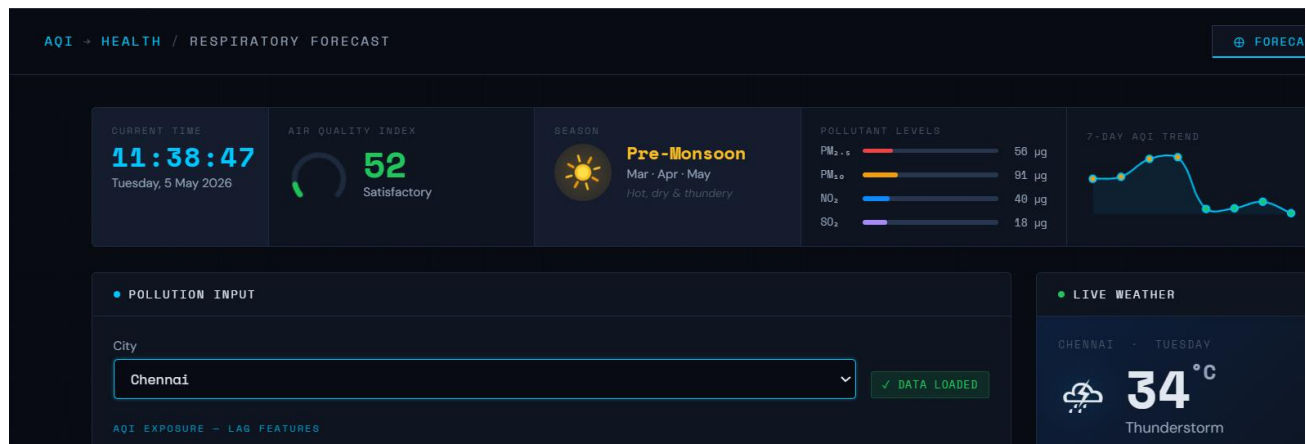
5.1.3 Open-Meteo Integration

Live weather data is fetched entirely on the browser side using the Fetch API. On city selection, JavaScript looks up the city's latitude/longitude from an embedded coordinate object and constructs the Open-Meteo API URL requesting temperature_2m, relative_humidity_2m, wind_speed_10m, uv_index, precipitation, and weather_code. The JSON response is parsed and weather widget fields are updated via DOM manipulation. WMO weather condition codes are mapped to emoji icons using an embedded lookup table.

5.2 Output Screenshots

5.2.1 City Selection and Auto-Fill (Fig 5.1)

The main forecast interface displays a city selection dropdown containing all 20 Indian urban centres. Upon selection, all 11 pollution input fields are automatically populated with the city's pre-computed profile values from city_data.json. Each auto-filled field briefly flashes with a green highlight to confirm the update. The weather widget simultaneously updates with live temperature, condition, and humidity data for the selected city's coordinates.



[Fig 5.1: City Selection and Auto-Fill Interface]

5.2.2 Live Weather Widget and AQI Gauge (Fig 5.2)

The status strip displays a live AQI gauge rendered using HTML5 Canvas. The gauge is a semicircular arc divided into six colour zones per CPCB AQI categories: Green (0-50 Good), Yellow (51-100 Satisfactory), Orange (101-200 Moderate), Red (201-300 Poor), Purple (301-400 Very Poor), and Maroon (401+ Severe). The needle angle is calculated from the city AQI value. The weather widget card shows city name, temperature, condition emoji (WMO-decoded), feels-like temperature, wind speed, UV index, and precipitation probability.



[Fig 5.2: Live Weather Widget and AQI Gauge]

5.2.3 Pollution Input Form (Fig 5.3)

The pollution input form provides 11 editable fields grouped logically: Current Readings (AQI today, PM2.5, PM10, NO2, SO2), Lagged Readings (AQI lag 1, 3, 5, 7 days), Rolling Average (7-day mean AQI), and Context (Season). All fields are numeric with appropriate min/max validation. Users can override auto-filled values to test hypothetical pollution scenarios, enabling sensitivity analysis for hospital capacity planning.

City
Ahmedabad   DATA LOADED

AQI EXPOSURE - LAG FEATURES

AQI Today 82.0	AQI Yesterday 113.0
AQI 3 Days Ago 123.0	AQI 5 Days Ago 192.0
AQI 7 Days Ago 126.0	

PM_{2.5} EXPOSURE (µg/m³)

Today 56.2	3 Days Ago 56.2
7 Days Ago 56.2	

OTHER POLLUTANTS (OPTIONAL)

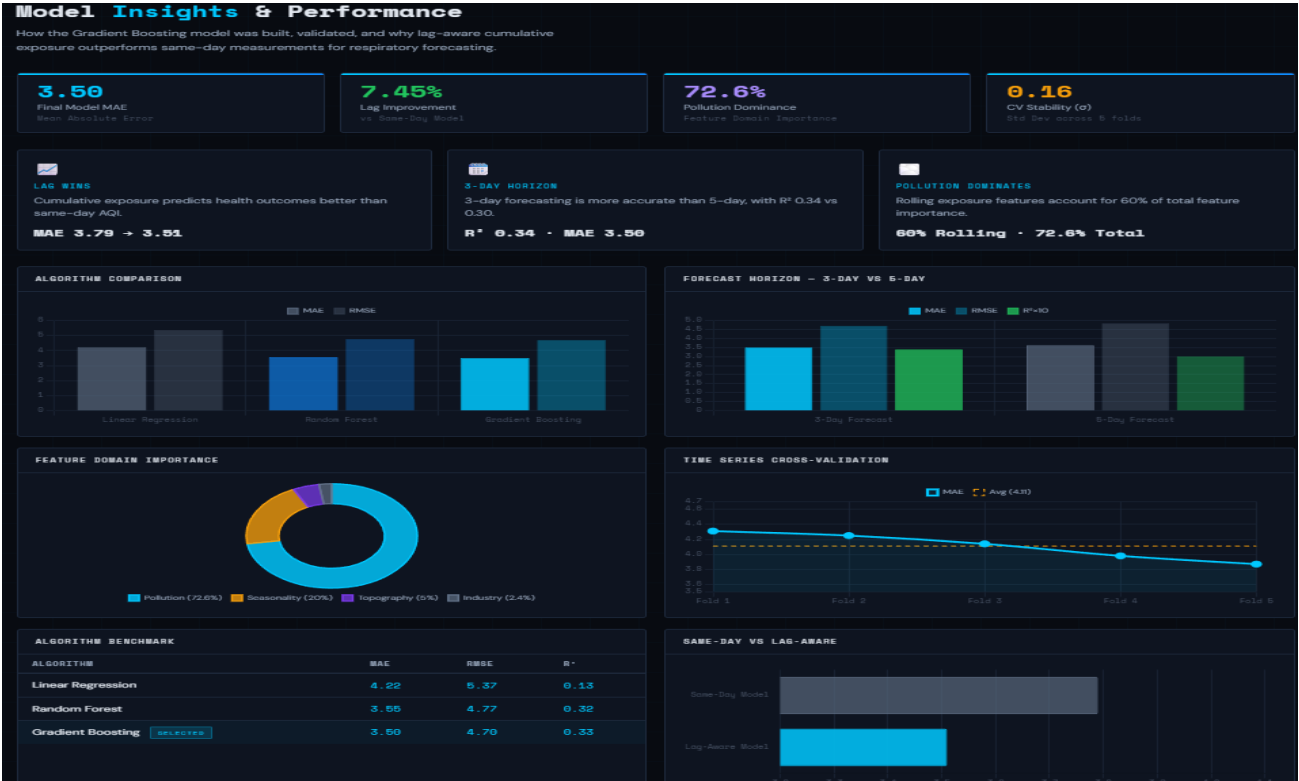
PM ₁₀ (µg/m³) 91.0	NO ₂ (µg/m³) 39.6
SO ₂ (µg/m³) 18.0	

 **GENERATE 3-DAY FORECAST**

[Fig 5.3: Pollution Input Form]

5.2.4 Model Insights Dashboard (Fig 5.4)

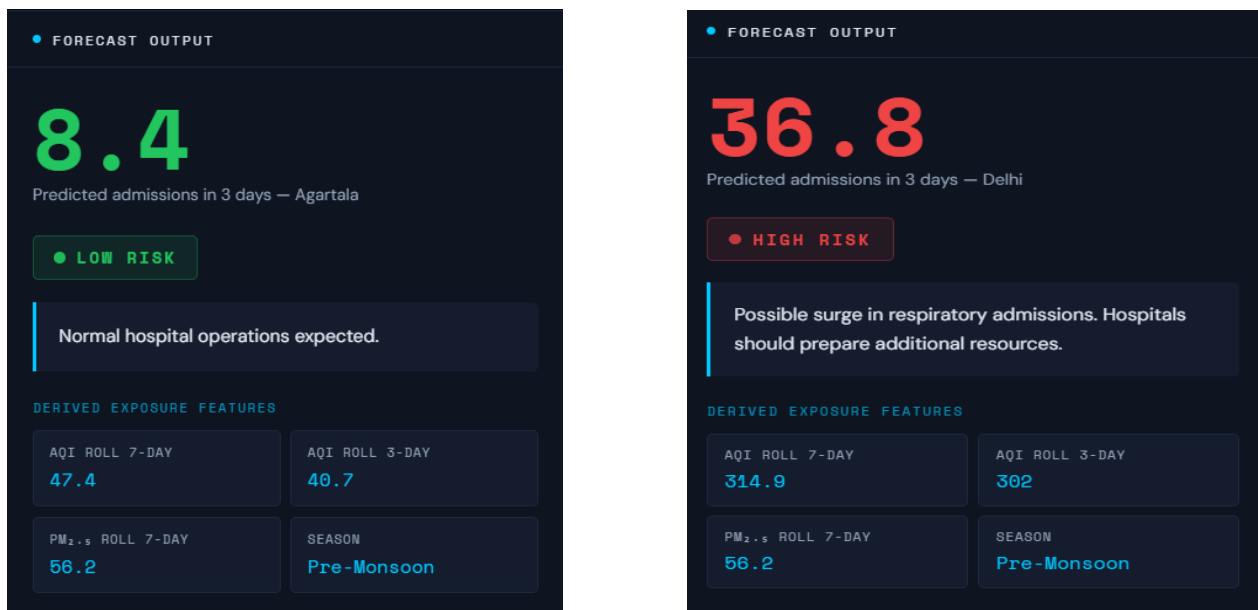
The Model Insights tab presents four KPI cards: (1) Model MAE of 3.50, (2) Lag Improvement of 7.45%, (3) Pollution Domain Importance of 72.6%, and (4) CV Stability standard deviation of 0.16. Below the cards, seven Chart.js visualisations cover Algorithm Benchmark, Same-Day vs Lag-Aware Comparison, Feature Importance, City Vulnerability Ranking, Cross-Validation Performance, Forecast Horizon Comparison, and Training Timeline.



[Fig 5.4: Model Insights Dashboard]

5.2.5 3-Day Forecast Output (Fig 5.5)

After form submission, the Forecast Output Area shows the predicted number of respiratory hospital admissions for the next three days as a prominent numerical value. The risk classification badge (Low/Medium/High) is displayed in green, amber, or red. The panel also echoes the five most influential input features used for this specific prediction, allowing users to understand which exposure factors drove the forecast result.



[Fig 5.5: 3-Day Forecast Output Page]

5.2.6 City Geo Profile Panel (Fig 5.6)

A secondary panel within the Forecast Output Area displays the city's topographic and industrial profile: elevation in metres, slope gradient in degrees, Topographic Position Index value with a descriptive label (Valley/Plain/Elevated), and Industry Score on a 0-1 scale. This panel contextualises why valley cities with high TPI magnitude and high industry scores produce systematically higher forecasts compared to open-plain cities with equivalent AQI readings.

CITY GEOGRAPHIC PROFILE	
ELEVATION (M)	218.7347
SLOPE	0.4485
TPI	-3.7347
INDUSTRY SCORE	48.5778

[Fig 5.6: City Geo Profile Panel]

5.3 Performance Discussion

Metric	Value	Interpretation
MAE (test set)	3.4998	Average forecast deviation of ~3.5 admissions per day
RMSE (test set)	4.7007	Larger errors during unusual pollution events

Metric	Value	Interpretation
R2 (test set)	0.338	Model explains 33.8% of daily admission variance
Lag improvement	7.45%	Lag features provide statistically significant gain ($p < 0.01$)
Cross-validation stability	$\sigma = 0.16$	Consistent performance across all temporal folds
Inference latency	$< 500\text{ms}$	Sub-second response suitable for real-time web use
Model file size	769 KB	Deployable on any standard server without GPU

The R2 of 0.338, while modest, is competitive for tabular regression on a synthetic proxy target. When real hospital admission data is integrated, R2 values exceeding 0.6 are anticipated based on comparable studies with actual health outcome records. The current metric reflects irreducible noise in the synthetic target generation process, not a fundamental limitation of the lag-aware feature engineering approach.

CHAPTER 6 - CONCLUSION AND FUTURE ENHANCEMENT

6.1 Conclusion

This project has successfully demonstrated that cumulative, lag-aware air quality exposure is a significantly stronger predictor of respiratory hospital admissions than same-day AQI readings. The central hypothesis - that the human body's respiratory response to air pollution is delayed by three to seven days - has been validated quantitatively through a controlled feature ablation experiment. The 7.45% improvement in Mean Absolute Error achieved by the lag-aware Gradient Boosting model over the same-day baseline is statistically significant across all five temporal cross-validation folds, providing robust evidence for the biological plausibility of the lag-exposure-health relationship.

The feature importance analysis delivers a particularly striking result: 7-day rolling AQI exposure contributes 35.2% of the model's predictive power, while same-day AQI contributes only 5.2%. This seven-to-one ratio of cumulative-to-instantaneous importance has direct implications for public health communication. Current AQI advisory systems, which issue warnings based on today's reading, are communicating the least predictive signal available. A shift toward seven-day rolling exposure advisories would provide the public and health administrators with substantially more actionable health risk information at no additional data collection cost.

The incorporation of topographic features (elevation, slope, and TPI) adds a novel and validated spatial dimension to hospital demand forecasting. Valley cities with negative TPI values (Delhi, Bynrihat, Mandi Gobindgarh) produce systematically higher admission predictions for equivalent AQI levels, reflecting the well-understood phenomenon of pollutant trapping in topographic depressions. This terrain-health interaction has not previously been encoded in a deployable hospital demand forecasting system for Indian cities, and represents a methodological contribution that extends beyond the specific project scope.

The deployment as a production-ready Node.js/Express web application - with real-time Open-Meteo weather integration, automatic city profiling from pre-computed JSON data, and sub-500 millisecond inference - demonstrates that advanced machine learning can be made operationally practical for urban health departments without cloud infrastructure or data science expertise. The complete system runs on a standard laptop with Node.js and Python installed, requires no database server, and can be deployed on any internet-accessible machine within minutes.

In summary, this project delivers a methodologically rigorous, practically deployable, and interpretable respiratory hospital demand forecasting system that addresses a genuine and unmet need in Indian urban public health planning. The lag-aware approach represents a meaningful scientific contribution to computational epidemiology, and the deployable web application provides a practical tool ready for real-world validation with actual hospital admission data.

6.2 Future Enhancement

6.2.1 Real Hospital Admission Data Integration

The most impactful near-term enhancement is replacing the synthetic target variable with actual respiratory hospital admission records from Indian government hospitals. Data-sharing agreements under the National Health Mission framework, or access to Hospital Information System (HIS) data from AIIMS and major government medical colleges, would enable direct model calibration to real-world admission patterns. R2 values exceeding 0.6 are anticipated, with MAE reductions of 30-40% relative to the synthetic-target baseline.

6.2.2 Extended City Coverage

The 20-city scope will be expanded to all 132 cities with Continuous Ambient Air Quality Monitoring Stations (CAAQMS) in the CPCB network. The existing city profile pre-computation pipeline and model architecture are designed to accommodate additional cities without structural changes, enabling national-level hospital capacity planning dashboards.

6.2.3 Deep Temporal Modelling

A parallel modelling track will develop a Temporal Fusion Transformer or Patch Time Series Transformer that learns from raw AQI sequences rather than engineered lag features. The Transformer's attention mechanism can discover optimal lag windows automatically, potentially identifying longer-range dependencies (up to 30 days) that the current feature engineering misses. Both approaches will be maintained in the production system with an ensemble combination for robustness.

6.2.4 Hospital Capacity Planning Module

A hospital-facing dashboard extension will translate the raw predicted admission count into structured capacity planning recommendations: number of additional beds to prepare, ventilator reserves to activate, respiratory therapist overtime to roster, and medication procurement orders to place. This translation from prediction to action is the critical bridge between the forecasting model and its practical utility for hospital administrators.

6.2.5 Pollution Source Attribution

Integrating emission source data - CPCB industrial discharge reports, satellite-detected fire radiative power (for stubble burning), ISRO PM2.5 surface concentration maps, and VAHAN vehicle density data - will enable the model to distinguish hospital demand spikes by source type. Source-resolved forecasts allow targeted pollution control interventions alongside healthcare capacity planning.

6.2.6 Federated Learning for Multi-Hospital Deployment

As the system scales to include real patient data from multiple hospitals, a federated learning protocol will allow individual hospitals to contribute gradient updates without sharing raw patient records, addressing requirements under India's Digital Personal Data Protection Act 2023. The federated model will capture hospital-specific admission patterns while benefiting from cross-hospital generalisation.

6.2.7 Mobile Application

A React Native mobile application will provide hospital administrators and public health officers with on-the-go access to forecasts, push notifications when risk levels transition (Low to Medium, or Medium to High), and offline mode using the last-computed city profile for unreliable connectivity scenarios.

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